12TH BIOLOGY · CHAPTER 23 · CHAPTER TEST

Pharmacology

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Biology	**Total Marks:** 26
Chapter: 23	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. The first step in drug discovery is:

(a) Lead optimization (b) Preclinical testing
(c) Hit identification (d) Target identification

2. Penicillin is highly effective against:

(a) Gram-negative bacteria (b) Gram-positive bacteria
(c) Streptomyces (d) Penicillium notatum

3. Tetracycline binds to which ribosomal subunit of bacteria?

(a) 30S (b) 50S
(c) 60S (d) 80S

4. The drugs derived from sulfanilamide are:

(a) Penicillin (b) Fluoroquinolones
(c) Sulfonamides (d) Tetracycline

5. Retroviruses replicate by using the enzyme:

(a) DNA polymerase (b) Reverse transcriptase
(c) Ligase (d) Lysozyme

6. Phase 1 clinical trials are carried out on:

(a) Animals (b) 100–300 patients
(c) 20–100 healthy volunteers (d) 1,000–3,000 patients

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Define pharmacology.
ii. What are lead compounds?
iii. Write the side effects of penicillin.
iv. Write the mode of action of tetracycline.
v. Differentiate between antiviral and antiretroviral drugs.
vi. What are the advantages of monoclonal antibodies?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Describe the stages of the drug developmental process.
(b) Describe penicillin and its mode of action.

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